LEF MODEL · CHAPTER 23

Chapter 23: The Planetary Rings — the Source of the Material, the Threshold Day of 13.4 Hours and Saturn's Equilibrium Flow

Based on Paper 12 — 12. The Rings: A ring breaks, grinds and returns (Parts A–D: the two lines, the two feeding paths, the threshold day of 13.4 h, Saturn's equilibrium and the escape of water)

01 · THE MYSTERY

The mystery: where does the ring material come from, why does only Saturn have enormous rings, and why don't rings build moons?

In Chapters 21–22 we saw that the moons in the solar system are born directly from their parents on the corotation line. In Paper 12 the LEF model extends this principle to **planetary ring systems** and sets an unambiguous rule: **a ring is not a construction site for moons but a mechanism that breaks, grinds and returns material to the parent!** From the observations of Cassini, Voyager and the telescopes, four central questions arise:

1. **The extreme differences between the rings of the solar system (A.1–A.3):** why are Saturn's rings huge (1.54×10^{19} kg) and bright (95%–99% pure water ice), Jupiter's rings made of faint dust, Uranus' and Neptune's rings narrow and dark — while around small bodies (Haumea, Chariklo and Quaoar) dense rings orbit *far beyond the Roche limit* without coalescing into moons?

2. **The paradox of the young age of Saturn's rings (D.3):** Cassini measured "ring rain" falling onto Saturn at a rate of **432 to 45,000 kg per second** — a rate that would have emptied the rings completely within 11 million to 1.1 billion years. How do the rings exist today?

3. **The universal condition for a massive ring (B.5):** why, of all the planets in the solar system, did only **Saturn today and young Mars** develop a massive ring directly from the ground?

4. **Where did the angular momentum of Uranus and Neptune go (D.4)?** The moons of Uranus and Neptune carry only 1% and 0.002% of the angular momentum these planets lost — what carried the rest away?

02 · AN IMAGE

An image: a net and balls, a grinding funnel, and a water fountain in equilibrium

Four tangible images explain the physics and the cycle of rings in the solar system:

- **A net and balls — the corotation line inside the Roche limit (B.5):** to capture material as a massive ring around a planet, two things are needed at the same time: **a net in the right place** and **balls thrown into it**. When material leaks from the depths of the planet to the corotation line (a_c), if the line lies *inside* the Roche limit ($a_c < a_R$), the differential tidal forces prevent it from coalescing into a moon and trap it as a ring. Dividing the two equations cancels the planet's density completely and yields **a universal threshold day, identical for every body in the universe:** $P^* \approx 13.4$ **hours!**
- **A viscous funnel that grinds lumps and returns dust home (C.1–C.4, D.1):** inside a ring disc, inner particles move faster than outer particles. Inelastic collisions turn orbital energy into heat while conserving the total angular momentum — so most of the material loses energy and flows inward to the parent, while a minority is pushed outward (that is why Saturn's ring A ends exactly at the Roche limit for ice at 135,300 km, and why Quaoar's ring at 2.4 Roche limits does not coalesce into a moon either!).

- **A fountain pool in which the feeding from below equals the rain from above (Saturn's equilibrium, D.3):** Saturn's rings are not a static remnant formed recently, but **a continuous flow in dynamic equilibrium**: the inner pump leaks water to the corotation line (1.92 R) at exactly the same rate at which the ring rain brings water down into the atmosphere (a cumulative cost of up to 1.1 × 10^{-5} of Saturn's mass over 4.5 billion years). The "young" appearance comes from the constant turnover of the material!
- **A mechanical pump lifting water from a shallow well (the escape of water from Uranus and Neptune, D.4):** Uranus and Neptune are far from the Sun and cold, yet they lost water amounting to **270 times the mass of all their moons** (2.5 × 10^{24} kg at Uranus and 3.2 × 10^{24} kg at Neptune) — not because of thermal evaporation, but because of **mechanical lifting to the corotation line** in the fast rotation (7–8 hours), through a thin gas lid (10%–20%) and a shallow gravity well ($v_{\mathrm{esc}} = 21.3$–23.5 km/s against 59.5 km/s at Jupiter)!

03 · SCIENCE AND MODEL

Science and model: the two families of rings, the two feeding paths and the matrix of conditions (Parts A–D)

Parts A–D of Paper 12 present quantitative proofs of four fundamental laws of ring systems:

1. **Division into two dynamic families relative to the corotation line (A.3):**
    - 2. **The family of the giants (0.50–$1.41\ a_c$):** the rings of Jupiter (0.57–$1.41\ a_c$), Saturn (0.60–$1.23\ a_c$), Uranus (0.51–$0.62\ a_c$) and Neptune (0.50–$0.75\ a_c$) sit below the corotation line or cross it, mostly inside the Roche limit; the part below the line falls inward to the parent (as also shown by the "spokes" in Saturn's ring B, which appear at 102,000–112,000 km, exactly up to the corotation line at 111,600 km!).
    - 3. **The family of the small bodies (1.25–$2.18\ a_c$):** the rings of Haumea ($2.07\ a_c$), Chariklo ($2.10; 2.18\ a_c$) and Quaoar ($1.25; 2.01\ a_c$) sit far *above* the corotation line and cluster around the radius of **the 1:3**

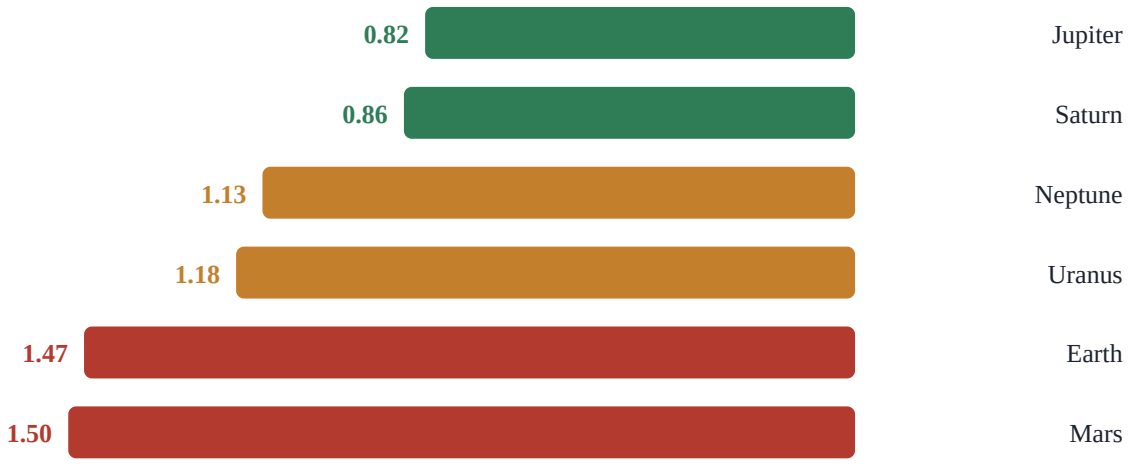
spin–orbit resonance ($a_{1:3} = 3^{2/3} a_c \approx 2.08\, a_c$), where the tide pushes material outward and the resonance traps it!

4. **Ruling out interplanetary dust as a source and defining the two feeding paths (B.2–B.4):** the chemical cleanliness of Saturn's rings (95%–99% pure water ice in an equatorial disc 10–30 m thick), the absence of shared angular momentum for isotropic space dust (which pollutes the dark ring C from outside and does not build it), and the precise correlation with the thickness of the atmospheric lid prove that all the ring material comes from the parent by two paths: **(a) direct feeding from the ground** below the corotation line (Saturn's rings D, C, B and the ancient ring of Mars), and **(b) feeding through active original moons** (Enceladus feeding ring E with geysers, Jupiter's four inner moons feeding dust, and Mab feeding Uranus' μ ring — while captured moons such as Phoebe are excluded).

5. **Applying the two conditions for a massive ring to all the planets (B.5):** Saturn today (P = 10.6 h < 13.4 h, $a_c = 1.92R < a_R = 2.24R$ + a leaking lid) and young Mars (P = 7–9 h < 13.4 h, $a_c = 1.74\text{–}2.05R < a_R = 2.68R$ + a volcano on a bulge) are the only ones that meet both conditions together. Jupiter (9.9 h) is fast enough but its gas lid is completely sealed; the Earth at birth (5.8 h) dried out too fast; and Uranus and Neptune today (17.2 and 16.1 h) are too slow ($a_c > a_R$) and frozen.

`The U-curve of the corotation line relative to the Roche limit ($x_c/x_R$, Figure B.3)`

In the middle of the solar system, at Jupiter and Saturn with their thick lids, the corotation line stays inside the ring zone ($x_c/x_R = 0.82$ and 0.86); while at both ends, where mass escaped through the line and slowed the rotation, the line left the ring zone — a little at Uranus and Neptune (1.18 and 1.13, where narrow rings remained), a lot at Mars (1.50, where the ring disappeared) and at the Earth (1.47), and up to 22 and 57 times at Mercury and Venus!



The ratio of the corotation line to the Roche limit (x_c/x_R) today: below 1 the line is inside the ring zone; Mercury (22) and Venus (57) are far off the scale.

04 · THE MECHANISM IN WORDS

The mechanism in words: deriving the universal threshold day (13.4 hours) and Saturn's equilibrium

Parts B.5 and D.1–D.4 of Paper 12 show how the geometry of orbits and the conservation laws dictate the existence, age and composition of planetary rings:

1. **Deriving the universal threshold day $P^* \approx 13.4$ hours (§B.5):** the corotation line (a_c) depends on the parent's mass and its rotation rate, while the Roche limit for ice particles (a_R) depends on the parent's radius and its mean density. When a_c is divided by a_R, the parent's mass, radius and density **cancel out completely**! The result is that the critical rotation day below which the corotation line lies inside the Roche limit is identical for every planet, moon or asteroid in the universe: $P^* \approx 13.4$ **hours** (for ice at a density of $\rho_{\text{ice}} = 900$ kg/m³). So the strong gravity of Jupiter or the weak gravity of young Mars has no effect on the threshold itself — only the rotation rate (P < 13.4 hours) and the leakage of material from the ground decide.

2. **Viscous spreading — why the ring is a funnel that returns material home (§D.1):** inside the ring disc, inner particles move faster than outer ones (differential Keplerian motion). Inelastic collisions turn orbital energy into heat while conserving the total angular momentum. To satisfy both

conservation laws together, the disc spreads: **most of the ring's mass loses energy and flows inward to the parent**, while a minority is pushed outward, carrying with it the excess angular momentum.

3. **Solving the paradox of the age of Saturn's rings (§D.3):** Cassini measured material falling from the rings onto Saturn in two parallel channels — neutral rain to the equator at a rate of **4,800–45,000 kg/s** (Waite et al. 2018) and ionised rain along the magnetic field lines at a rate of **432–2,870 kg/s** (O'Donoghue et al. 2019). Without feeding, the mass of the rings (1.54×10^{19} kg) would empty within 11 million to 1.1 billion years. In the LEF model, the ring is **a constant flow in dynamic equilibrium**: water leaking from the inner envelope renews it from below at the same rate at which the rain empties it from above — at a cumulative cost of **only 10^{-7} to 1.1×10^{-5} of Saturn's mass** over 4.5 billion years!

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The ratio between the corotation line a_c and the Roche limit for ice a_R, and the universal threshold day P^*, are formulated as follows (§A.2, §B.5):

$$a_c = \left(\frac{GM}{\Omega^2}\right)^{1/3}$$

$$a_R \approx 2.46\, R \left(\frac{\rho}{\rho_{\text{ice}}}\right)^{1/3}$$

$$\frac{x_c}{x_R} = \left(\frac{P}{P^*}\right)^{2/3}$$

$$P^* = 2\pi \sqrt{\frac{3 \cdot 2.46^3}{4\pi G \rho_{\text{ice}}}} \approx 13.4 \text{ hours}$$

Where $\rho_{\text{ice}} = 900$ kg/m³, one gets $a_c < a_R$ if and only if P < 13.4 hours. Combined with the second condition (active leakage of water and material from the ground or the envelope), a sharp division of the whole solar system follows.

06 · THE SIGNATURE

The signature in reality: the test of $P^* = 13.4$ hours and the angular-momentum budget of the ice giants

Comparing the Cassini and Voyager spacecraft data with the conservation equations confirms two prominent quantitative findings (§B.5, §D.4):

- **Matching the massive-ring condition across eight planetary states (§B.5):** only Saturn today (P = 10.6 h, $x_c/x_R = 0.86$ + a leaking lid) and young Mars (P = 7–9 h, $x_c/x_R = 0.65$–0.76 + a volcano on a bulge) meet both conditions for a massive ring directly from the ground.
- **The 270-fold gap in the mass of water that escaped from Uranus and Neptune (§D.4):** slowing Uranus from 8.3 to 17.2 hours removed an angular momentum of 1.3 × 10^{36} kg·m²/s, of which the moons hold only 1% (1.4 × 10^{34}). The water that escaped through the corotation line (2.5 × 10^{24} kg, about 2.8% of Uranus' mass) weighed **270 times the mass of all its moons together** (9 × 10^{21} kg), and at Neptune (3.2 × 10^{24} kg, about 3.1% of the planet's mass) it carried 99.998% of the momentum lost!

The lab box: the matrix of massive-ring conditions ($P^* = 13.4$ h), the ring-rain rate at Saturn and the angular-momentum budget (Paper 12, Parts A–D)

Planet and period	Day P (h)	Corotation line / Roche limit (R)	Condition 1 (P < 13.4 h)	Condition 2 (leakage from the ground)	The dynamic outcome
Saturn today	**10.6**	**1.92 / 2.24 ($x_c/x_R = 0.86$)**	**Yes**	**Yes (the lid leaks)**	**A massive ring directly from the ground**

Young Mars	**7–9**	**1.74–2.05 / 2.68**	**Yes**	**Yes (a volcano on a bulge)**	**An ancient ring existed**
Jupiter today	9.9	2.29 / 2.80 ($x_c/x_R = 0.82$)	Yes	Almost zero (thick lid)	A thin dust ring from moons only
The Earth at birth	5.8	2.29 / 4.02	Yes	No (dried out fast)	No ring; the Moon was built on a pear
Uranus / Neptune today	17.2 / 16.1	3.26/2.76 ; 3.39/3.00	**No**	No (the water escaped and it froze)	Narrow remnants and moon rings
Mars / Earth today	24.6 / 23.9	6.03/4.02 ; 6.62/4.49	**No**	No (the crust sealed)	The ring disappeared / no ring

Ring-rain rate at Saturn (kg/s)	Emptying time without feeding	Total feeding required over 4.5 Ga	Fraction of Saturn's total mass
432 (ionised rain — lower bound)	1.1 billion years	4 ring masses	1×10^{-7}
2,870 (ionised rain — upper bound)	170 million years	26 ring masses	7×10^{-7}
4,800 (neutral rain — lower bound)	100 million years	44 ring masses	1.2×10^{-6}
45,000 (neutral rain — upper bound)	**11 million years**	**415 ring masses**	**1.1×10^{-5} (within mass conservation!)**

07 · NEXT

The door ahead

In this chapter we derived **the universal threshold day of $P^* \approx 13.4$ hours** for a massive ring, saw how Saturn's rings exist as a constant equilibrium flow, and proved that the mechanical escape of water through the corotation line carried 99%–99.998% of the angular momentum lost by Uranus and Neptune.

In the next chapter — **Chapter 24: The Ring as a Grinding Conveyor, the Signatures on Moons and Planets, and the Atacama Block Test** (from Paper 12 — Parts E–M) — we will examine how a ring sorts lumps in collisions and produces "propellers", why an ice ring builds equatorial ridges on Pan and Atlas while a

silicate ring wears down Deimos, how the Atacama rock hypothesis was ruled out at Curiosity in April 2026, and we will gather the 5 falsification criteria of Part L!

Chapter 24

The Ring as a Grinding Conveyor, the Signatures on Moons and Planets, and the Atacama Block Test →

← Chapter 22 · Prologue

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